

Eduardo Aguilar-Bejarano

Machine Learning for Chemistry & Materials Sciences

PhD, Chemistry & Artificial Intelligence · University of Nottingham

✉ ed.aguilar.bejarano@gmail.com [edaguilarb.github.io](https://github.com/edaguilarb) [in edaguilar00](https://www.linkedin.com/in/edaguilar00)

[EdAguilarB](#) [ID 0009-0007-6251-2350](#) [Google Scholar](#)

PROFILE

Chemist and machine-learning researcher developing data-driven approaches to fundamental problems in chemistry, materials science, and engineering. I am particularly interested in incorporating domain knowledge, from physical chemistry to biology, into ML models, and in using interpretable models to turn predictions into testable structure–property hypotheses. My work spans asymmetric and organometallic catalysis, polymers, and lipid nanoparticles, developed in close collaboration with experimental groups and validated prospectively on newly synthesized systems. I also build AI workflows and open-source tools that meet the practical needs of experimental scientists, accelerating experimentation while reducing research costs and environmental impact. I believe AI should help us solve cutting-edge problems faster without sacrificing a fundamental understanding of the phenomena behind them.

Interests

Asymmetric & organometallic catalysis · representation learning · interpretable AI models · generative molecular design · benchmarking & foundation models · open-source scientific software

EDUCATION

2022 – 2026

PhD in Chemistry & Artificial Intelligence

Nottingham, UK

University of Nottingham — AI Doctoral Training Centre

Thesis: *Interpretable Graph Neural Networks for the Discovery of Structure–Activity Relationships*

Supervisors: Prof. Simon Woodward, Prof. Graziela Figueredo, Prof. Ender Özcan · Defended September 7, 2026

2018 – 2022

BSc (Hons) Chemistry

San José, Costa Rica

Universidad de Costa Rica

GPA 9.23 / 10.0 · Graduated with distinction (academic excellence)

Relevant coursework: Bio-organic Chemistry, Bio-inorganic Chemistry, Computational Tools for Physical Chemistry, Scientific Computation with Python, Introduction to Genomics and Metagenomics Data Processing, Automated Learning with Python.

RESEARCH EXPERIENCE

Nov 2024 –
Present

Data Scientist (part-time)

Nottingham, UK

Digital Research Service, University of Nottingham

EPSRC Large Grant: Biomaterials for Medical Devices AI

- Lead modeler for an antibiofilm polymer-discovery program: trained ensemble GNNs on ~500 polymer structures against high-throughput biofilm assays, in iterative collaboration with experimental microbiologists.
- Delivered **LCNet**, the neural-network model behind the first quantitative Q-Cryo-OrbiSIMS method for lipid-nanoparticle characterization (with MIT's Anderson Lab); [ChemRxiv preprint](#).
- Architect and delivery lead for **Helix**, an open-source framework for reproducible, interpretable ML on tabular scientific data, adopted across chemistry, materials, and biology at Nottingham.

Jun – Aug 2026

Visiting Researcher

Cambridge, MA, USA

Anderson Lab, Massachusetts Institute of Technology

Generative and interpretable models for molecular design

- Developed conditional generative models (diffusion and VAE latent-space optimization) for the design of novel lipid structures, coupling generative sampling with property-prediction models to steer candidates toward target profiles.
- Built interpretable models to prioritize lipid nanoparticles, extending prior composition-analysis work toward model-guided exploration of chemical space.

Oct 2022 –
Sep 2026

PhD Researcher, Chemistry & Artificial Intelligence

Nottingham, UK

University of Nottingham

AI Doctoral Training Centre · Woodward, Figueredo & Özcan Research Groups

- Showed that graph encoding of planar and axial chirality predicts enantioselectivity as accurately as models built on expert-derived descriptors, without requiring them, and transfers to unrelated reaction classes (catalytic asymmetric dearomatization, N,S-acetal formation).
- Prospective validation with synthetic collaborators: predictions tested against 52 newly synthesized Rh-catalyzed 1,4-addition reactions (Lam Group, Nottingham) and 19 new Pd-catalyzed DAAA reactions (Guiry Group, University College Dublin).
- Established that scaffold-based splits, not random splits, predict the real-world error of asymmetric-catalysis models — a benchmark across fingerprints, descriptors, pretrained chemical foundation models, and GNNs.
- Showed that GNN-based data-quality workflows can flag mislabeled entries in published asymmetric-catalysis datasets.
- Developed a 3D-masking explainability method for physics-based (crystal-convolution) GNNs predicting interstitial-alloy properties, and trained machine-learned interatomic potentials.

Advisor: Prof. William Zamora-Ramírez

- Built classical ML models (multiple linear regression, SVM, random forest, gradient boosting) to predict partition coefficients (logP, water/toluene) and tautomerization constants of drug-like small molecules, including database curation, molecular-representation design, and feature selection.

PUBLICATIONS

Complete, up-to-date list on [Google Scholar](#) and [ORCID](#).

PEER-REVIEWED JOURNAL ARTICLES

- [5] **Aguilar-Bejarano, E.**; Lea, D.; Sivakumar, K.; Mase, J. M.; Omidvar, R.; Li, R.; Kettle, T.; Mitchell-White, J.; Alexander, M. R.; Winkler, D. A.; Figueredo, G. Helix 1.0: An Open-Source Framework for Reproducible and Interpretable Machine Learning on Tabular Scientific Data. *Patterns* **2026**, 7 (5), 101536.
[doi:10.1016/j.patter.2026.101536](#) · IF 10.8
- [4] **Aguilar-Bejarano, E.**; Galvin, D.; Rogers, D. M.; Özcan, E.; Woodward, S.; Guiry, P. J.; Figueredo, G. Benchmarking Molecular Representations and Machine Learning Algorithms for Asymmetric Catalysis: A Palladium-Catalysed Decarboxylative Asymmetric Allylic Alkylation Case Study. *J. Cheminform.* **2026**, 18 (1), 120.
[doi:10.1186/s13321-026-01236-z](#) · IF 7.9
- [3] **Aguilar-Bejarano, E.**; Arrieta, L.; Gutiérrez, M.; Özcan, E.; Woodward, S.; Figueredo, G.; Borge, J. I. Explainable GNN-Derived Structure–Property Relationships in Interstitial-Alloy Materials. *Phys. Chem. Chem. Phys.* **2025**, 27 (41), 22240–22250.
[doi:10.1039/D5CP02208H](#) · IF 3.0
- [2] **Aguilar-Bejarano, E.**; Özcan, E.; Rit, R. K.; Li, H.; Lam, H. W.; Moore, J. C.; Woodward, S.; Figueredo, G. Homogeneous Catalyst Graph Neural Network: A Human-Interpretable Graph Neural Network Tool for Ligand Optimization in Asymmetric Catalysis. *iScience* **2025**, 28 (3), 111881.
[doi:10.1016/j.isci.2025.111881](#) · IF 4.5
- [1] **Aguilar-Bejarano, E.**; Deorukhkar, V.; Woodward, S. Data Checking of Asymmetric Catalysis Literature Using a Graph Neural Network Approach. *Molecules* **2025**, 30 (2), 355.
[doi:10.3390/molecules30020355](#) · IF 5.1

PREPRINTS

- [P1] Roberts, J. W.; Kotowska, A.; **Aguilar-Bejarano, E.**; Collins, E.; Witten, J.; Howard, A.; Hook, A.; Winkler, D. A.; Alvey, K.; Figueredo, G.; Scurr, D. J.; Anderson, D. G.; Alexander, M. R. Quantitative Cryogenic Orbitrap Secondary Ion Mass Spectrometry for Structural Characterization of Lipid Nanoparticles. *ChemRxiv* **2026**.
doi:10.26434/chemrxiv.15001097

OPEN-SOURCE SOFTWARE

- Catalysis **HCat-GNet** — interpretable GNN for ligand and catalyst optimization in asymmetric catalysis; attributes predictions to chemically meaningful substructures.
 [EdAguilarB/hcatgnet](#) · Paper [2]
- Catalysis **AsymBench** — benchmarking suite for asymmetric catalysis; modular base classes for adding representations, models, and extrapolative splits.
 [EdAguilarB/AsymBench](#) · Paper [4]
- Polymers **PolyNet** — polymer property prediction: six GNN architectures, automated hyperparameter optimization, atom- and fragment-level explainability, and a GUI for non-computational users.
 [Biomaterials-for-Medical-Devices-AI/PolyNet](#)
- General ML **Helix** — reproducible, interpretable ML on tabular scientific data. Architect and delivery lead, including interpretation modules, GUI, and documentation. Adopted across chemistry, materials, and biology at Nottingham.
 [Biomaterials-for-Medical-Devices-AI/Helix](#) · Paper [5]
- Lipid NPs **LCNet / LNPQuantification** — neural-network model behind the first quantitative Q-Cryo-OrbiSIMS method for lipid-nanoparticle characterization (with MIT's Anderson Lab); Integrated-Gradients interpretation.
 [Biomaterials-for-Medical-Devices-AI/LipidLNPQuantification](#) · Preprint [P1]
- Materials **InterstitialAlloys** — explainable GNNs for interstitial-alloy property prediction.
 [EdAguilarB/InterstitialAlloys](#) · Paper [3]

HONORS & AWARDS

FELLOWSHIPS & SCHOLARSHIPS

- 2022 **AI Doctoral Training Centre Scholarship**, University of Nottingham
Full four-year PhD funding (tuition and stipend) through a competitive EPSRC program.
- 2022 **CeNAT-CONARE Research Fellowship**, National Center for High Technology, Costa Rica
US\$4,300 for the project “Cheminformatic and Quantum Mechanics Approaches for Quantitative Prediction of Tautomerism in Bioactive Molecules.”
- 2021 **GCURS Acceptance & Travel Award**, Rice University, USA

- 2019 – 2022 **Academic Excellence Scholarship**, Universidad de Costa Rica
Annual award for a prior-year GPA of 9.0/10 or above.
- PRIZES & DISTINCTIONS**
- 2024 **Christopher J. Moody Synthesis & Catalysis Poster Prize**, School of Chemistry, University of Nottingham
- 2024 **Second Place, PGR Symposium Poster Contest**, University of Nottingham
- 2022 **Bachelor of Science with Honors**, Universidad de Costa Rica
Awarded for graduation with academic excellence.
- 2022 **Best Scientific Poster**, Chemistry Students Association, Universidad de Costa Rica
- 2021 **Profesor Distinguido (Distinguished Tutor)**, Organizing Committee of the Costa Rican Chemistry Olympiad
Recognized for the performance of my students and the outstanding overall results of Colegio Científico Costarricense, sede San Pedro, which placed 3rd among all participating institutions.
- OLYMPIADS**
- 2021 **Primer Puesto (First Place)**, Ibero-American University Organic Chemistry Olympiad, Universidad Nacional de Colombia
Also awarded best final exam and distinguished participant.
- 2017 **Gold Medal**, Costa Rican Physics Olympiad; **Honorable Mention**, Costa Rican Chemistry Olympiad, Universidad Nacional (UNA)

CONFERENCES & PRESENTATIONS

INVITED TALKS

- Mar 2024 *HCat-GNet: An Interpretable Graph Neural Network for Catalysis Optimization*
The GSK Prosperity Partnership Stevenage, UK
- Aug 2023 *Chemistry in the Artificial Intelligence Era*
School of Chemistry Seminar Series, Universidad de Costa Rica
San José, Costa Rica

CONTRIBUTED TALKS

- Jul 2026 *Molecular Representations in the Context of Homogeneous Asymmetric Catalysis*
(oral presentation)
School of Chemistry Postgraduate Symposium, University of Nottingham
Nottingham, UK
- Oct 2021 *Cheminformatic and Quantum Mechanics Approaches for Quantitative Prediction of Tautomerism in Bioactive Molecules* (oral presentation)
Gulf Coast Undergraduate Research Symposium (GCURS), Rice University
Houston, TX, USA

POSTERS

Jul 2024	<i>HCat-GNet: an Interpretable Graph Neural Network for Catalysis Optimization</i> School of Chemistry Postgraduate Symposium, University of Nottingham Nottingham, UK
Nov 2023	<i>Feature Identification in Molybdenum Carbides: Graph Neural Networks vs. Human Empirical Search</i> Faraday Community Poster Symposium, Burlington House London, UK
Sep 2023	<i>Feature Identification in Molybdenum Carbides: Graph Neural Networks vs. Human Empirical Search</i> Machine Learning for Atomistic Modelling Autumn School, Daresbury Laboratory Daresbury, UK
Sep 2022	<i>Development of a Novel Lipophilicity Descriptor from Toluene/Water Partition Coefficient Predictions</i> Sciences Week Scientific Poster Contest San José, Costa Rica
Apr 2022	<i>Machine Learning Methods to Determine the Toluene/Water Partition Coefficient as an Efficient Lipophilic Descriptor</i> University Week Scientific Poster Contest San José, Costa Rica

SUPERVISION & TEACHING

RESEARCH SUPERVISION

2024, 2026	Co-supervisor, summer research interns · Woodward & Figueredo Groups, University of Nottingham 2026: projects on prediction error / applicability-domain estimation and the open-source Helix tool. 2024: GNN-based curation of asymmetric-catalysis datasets (resulting in <i>Molecules</i> 2025, paper [1]).
2023 – 2024	Co-supervisor, MSc Machine Learning in Science · University of Nottingham 2024: three students, with Prof. Simon Woodward. 2023: one student (Veera Pedireddi), with Prof. Graziela Figueredo and Dr. Kristaps Ermanis. Deep-learning projects in the chemical sciences.

TRAINING & WORKSHOPS

Nottingham	Designed and delivered in-house Python and machine-learning training sessions for interdisciplinary research teams.
------------	---

TEACHING & TUTORING, COSTA RICA

Jul – Sep 2022	Programming, ML & Data-Analysis Tutor , CBIO3 Group Teaching assistant for Prof. William Zamora's Cheminformatics course.
Sep – Nov 2021	Organic & Analytical Chemistry Tutor , Campus Atlántico
Mar – Oct 2021	Chemistry Olympiad Tutor , Colegio Científico Costarricense, San Pedro Recognized as <i>Profesor Distinguido</i> by the Olympiad Organizing Committee (see Honors & Awards).
2021, 2022	General Chemistry Tutor for incoming students (March intakes)
2019 – 2021	Laboratory Teaching Assistant — seven courses Physical Chemistry I & II, Analytical Chemistry I, Organic Chemistry I, General Organic Chemistry I, Chemical Experimentation II, General & Qualitative Chemistry.
Mar – Jul 2019	Physics Reinforcement Tutor , Colegio Científico Costarricense, San Pedro
2018	Teaching Assistant to Prof. Darío Chinchilla (General & Biological Chemistry)

PROFESSIONAL SERVICE & MEMBERSHIPS

Peer review	Reviewer for <i>Journal of Cheminformatics</i> (Springer Nature), <i>Chemometrics and Intelligent Laboratory Systems</i> (Elsevier), <i>Engineering</i> (Elsevier), and <i>Archives of Computational Methods in Engineering</i> (Springer Nature).
2023 – Present	Royal Society of Chemistry — Associate Member
2021 – 2022	American Chemical Society — Member
2021 – 2022	Costa Rica ACS Student Chapter — Member

TECHNICAL SKILLS

Cheminformatics & Molecular Modeling

RDKit · SMILES/InChI · molecular representations (fingerprints, topological & physicochemical descriptors, molecular graphs, organometallic complexes) · QSAR/QSPR · Chemprop · OpenBabel · DataWarrior · PyMOL

Deep Learning & Machine Learning

PyTorch · PyTorch Geometric · GNNs (GCN, GAT, MPNN, GraphSAGE) · generative models (diffusion/flow-based, VAEs, latent-space optimization) · scikit-learn · XGBoost · transfer & self-supervised learning · ensemble methods · hyperparameter optimization (Ray Tune)

Interpretability & Foundation Models

GNExplainer · SHAP · Integrated Gradients · GradCAM · attention analysis · chemical foundation models (ChemBERTa, MolT5, UniMol) · LLM APIs (Anthropic Claude, OpenAI GPT)

Atomistic Modeling & Interatomic Potentials

Machine-learned interatomic potentials (MACE) · ASE · pymatgen · DScribe · quantum chemistry (Gaussian — QM continuum solvation)

Software Engineering & Computing

Python (advanced, OOP) · Git & code review · automated testing · CI/CD · Docker · Bash · R · Linux · HPC/SLURM · GPU/CUDA · Weights & Biases · Streamlit

Laboratory

HPLC · gas chromatography · atomic absorption · UV/Vis spectrophotometry · fluorescence spectrophotometry

Languages

Spanish (native) · English (fluent)

REFEREES

Prof. Simon Woodward — PhD primary supervisor

School of Chemistry, University of Nottingham

simon.woodward@nottingham.ac.uk

Prof. Graziela Figueredo — PhD co-supervisor & line manager

Digital Research Service, University of Nottingham

g.figueredo@nottingham.ac.uk

Prof. Ender Özcan — PhD co-supervisor

School of Computer Science, University of Nottingham

ender.ozcan@nottingham.ac.uk